Gauge Structure, QED, and Gravity from the Spectral Geometry of S^3: A Synthesis of the GeoVac QED/Gauge Arc*

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The Geometric Vacuum (GeoVac) framework labels the hydrogen bound states by the harmonics of the unit three-sphere S^3 via Fock's conformal projection. This synthesis assembles the five Group 5 papers—gauge structure, the vector-photon vertex and the periods of Dirac spectral sums on S^3, the fine-structure-constant observation, and a spectral-action gravity sketch—into one narrative around a single organizing distinction: *$U(1)$ and* SU(2) *Wilson structures can be built on the graph, the choice of* SU(2) *being natural because $S^3 =$ SU(2), while the couplings are calibration/observation, not derived.* On the scalar label graph the plaquettes (the 4-cycles) form a cycle basis, so SU(2) Wilson theory and its maximal-torus restriction, compact $U(1)$, are the exactly solvable two-dimensional theories, whose area law holds by factorization and carries no confinement information; on the cross-ℓ Dirac Rule-B graph compact $U(1)$ does not factorize and is compatible with compact electrodynamics under seven standard diagnostics, none of which discriminates a dimension [7]; the natural gauge reach is $U(1) \times$ SU(2) $\times$ SU(3) (an organizing reading, not a derivation), with SU(3) entering only as a pure-gauge theory whose natural matter coupling is obstructed. Of eight QED selection rules the scalar graph keeps one (triangle sparsity); the physical vector-photon vertex, computed exactly from Spin(4) representation theory, keeps triangle sparsity and Δm, satisfies the Ward–Takahashi identity, couples the ground state, and does not preserve the 1+6+1 partition of the vertex model [9]; the continuum S^3 Dirac spectral sums carry odd zeta values ($\zeta(3)$, $\zeta(5)$), the level-4 values Catalan G and $\beta(4)$ when the spectrum is restricted to one parity class, and, in the vertex model's three-loop chain sum, the depth-2 period $\zeta(5,3)$, through a depth-filtered period tower [6]. The fine-structure constant approximately satisfies (to 8.8×10^{-8} relative; an Observation) the cubic $\alpha^3 - K\alpha + 1 = 0$ with $K = \pi(B + F - \Delta)$ assembled from three spectral data of the unit S^3, each with an independently established spectral home; this is recorded strictly as an **Observation** (agreement 8.8×10^{-8}, twelve single-mechanism derivations eliminated), never as a derivation [2]; in a look-elsewhere search of the same kind the expected chance count at this precision is about ten. The gravity paper gives a convergence theorem for a discrete replica derivative on the spinor cone (one standard ingredient argued rather than proved), and records the two-term spectral action of the round sphere (Chamseddine–Connes; an alternative proof there) and a Newton constant obtained by two routes, whose agreement (at leading order in Λ on the black-hole cigar) is the induced-gravity statement of Susskind–Uglum and Frolov–Fursaev [13]. The discipline throughout matches the rest of the corpus: every load-bearing claim wears its tier, the structural results are foregrounded, and the couplings are positioned as the free, calibration side of the construction.

I. INTRODUCTION: NATURAL STRUCTURE, FREE COUPLINGS

The GeoVac program builds its discrete hydrogen lattice on the labels (n, l, m) of the Laplace–Beltrami operator on the unit three-sphere S^3, on which that operator acts as the diagonal $SO(4)$ Casimir; the operator is related to the Schrödinger equation by Fock's 1935 conformal projection [3]. (The lattice's graph Laplacian is not equivalent to it and does not converge to it, Paper 7.) The Group 5 arc asks what the *same* spectral geometry says about the rest of electromagnetism, the gauge sector, and—most speculatively—gravity. A single distinction organizes the answer, and it is the through-line of this synthesis.

The $U(1)$ and SU(2) *gauge structures are constructions the geometry makes natural, not consequences it forces; the $U(1) \times$ SU(2) $\times$ SU(3) saturation below is an organizing reading, not a derivation.* The three-sphere carries a canonical principal-bundle structure, the Hopf fibration $S^1 \to S^3 \to S^2$, and S^3 *is* the group manifold SU(2). These two facts are not GeoVac inputs; they are geometry. Reading the GeoVac graph through its incidence matrix, one can build an abelian and then a non-abelian Wilson lattice gauge theory on the same nodes and edges (a Wilson theory can be placed on any graph; what the geometry supplies is the natural choice of group), whose labels carry the Coulomb spectrum (§II; the spectrum sits on the labels as the diagonal $SO(4)$ Casimir, not in the edge Laplacian). The natural gauge reach saturates at $U(1) \times$ SU(2) $\times$ SU(3), and where it stops is itself structural: SU(3) admits only a pure-gauge Wilson theory, its natural matter coupling obstructed at the Clebsch–Gordan intertwiner level (a sharpening of the Coulomb/harmonic-oscillator asymmetry of Paper 24 [5]).

The couplings are not. The fine-structure constant,

* Group 5 synthesis in the Geometric Vacuum series

the QED multi-loop constants, and the cutoff function of the gravity sector are all on the other side of the ledger: calibration or observation, in the two-layer language of Paper 34 [10]. The framework's most precise number—the cubic's root matches α to 8.8×10^{-8} (K itself to 4.8×10^{-7})—is held explicitly at **Observation** tier, with no claim of derivation (§IV). This is the same discipline that runs through the quantum-chemistry and quantum-computing arcs: lead with the structural result, name the calibration boundary, and let the tier of each claim be visible in the prose.

Following the framework's positioning convention, the discrete graph topology and the continuous Hopf/spectral-action geometry are treated as *dual descriptions* connected by proven maps; no ontological priority is asserted for either, and the synthesis leads with concrete structural content rather than interpretive claims about the nature of electromagnetism or gravity.

The five papers form a loose dependency order. Paper 30 builds the gauge structure on the scalar and Dirac Rule-B label graphs. Paper 33 computes the physical vector-photon vertex on S^3 and scores eight selection rules on the graph, in a vertex model and with the physical vertex; Paper 28 determines the periods of the Dirac spectral sums, whose nested sums use that vertex model. Paper 2 reports the α observation. Paper 51 proves a convergence theorem for a discrete replica derivative and records the continuum spectral-action gravity results it serves, together with what did not work. The hydrogen Lamb-shift paper (Paper 36 [11]) uses no S^3 input in its number and is part of the precision group (Group 6).

II. WILSON GAUGE THEORY ON THE LABEL GRAPHS

a. The Hodge/Wilson dictionary on the graph (Paper 30 §4.2). With B the node–edge incidence matrix of the scalar label graph, the node Laplacian $L_0 = BB^\top$ is the scalar matter operator of the lattice gauge theory (it does *not* converge to $-\Delta_{S^3}$: it splits into $n_{\max}$ disconnected ℓ-blocks, and the Coulomb spectrum lives on the labels instead [3]), the edge Laplacian $L_1 = B^\top B$ has the cycle space as its kernel, and with the plaquette incidence B_2 the gauge-fixed (Feynman-gauge) photon propagator is the inverse of the Hodge-1 Laplacian $B^\top B + B_2 B_2^\top$. The dictionary is standard and holds on every graph. Collapsing each m_ℓ-multiplet at fixed (n, ℓ) to one vertex gives a weighted graph on the six sectors (n, ℓ) with Laplacian spectrum $\{0, 0, 0, 1, 3, 6\}$ (Paper 30 §4.3). Grouping each shell by the chiral SU(2) factor whose torus is the Hopf fibre instead of by ℓ halves the Casimir trace of the α observation, so that trace belongs to the rotation labelling, not to the fibration (Paper 2, scope section). None of the underlying mathematics is new (graph Hodge theory, Wilson lattice gauge theory, the Marcolli–van Suijlekom gauge-network spectral triple [14]); what Paper 30 adds is the identification of the label graphs' plaquette complexes and the measured Rule-B diagnostics. The edge Laplacian is an integer matrix, so its spectrum is algebraic (π-free; at $n_{\max} = 3$ its characteristic polynomial splits over $\mathbb{Q}(\sqrt{5})$), and no transcendental can hide in the gauge 1-forms themselves.

b. SU(2) *because* S^3 = SU(2) *(Paper 30 §3).* The rank-1 non-abelian case is special: the manifold *is* the gauge group. With links $U_e \in$ SU(2), plaquettes the graph's 4-cycles (the unit squares of the planar ℓ-blocks), and Wilson action $S_W = \beta \sum_P \big(1 - \frac{1}{2} \operatorname{Re} \operatorname{tr} U_P\big)$, the construction is a gauge-invariant, Haar-normalizable finite theory. The plaquettes form a basis of the cycle space at every $n_{\max}$, so after gauge fixing the plaquette holonomies are independent Haar variables: $Z(\beta) = [2e^{-\beta} I_1(\beta)/\beta]^{\beta_1}$ and every loop enclosing A plaquettes has $\langle W \rangle = (I_2/I_1)^A$ exactly, an area law by factorization that carries no information about confinement (Paper 30 Prop. 1). The second-order weak-coupling expansion produces the curl plaquette form $B_2 B_2^\top$ — *not* a multiple of L_1; it is the missing half that completes the edge Laplacian $L_1 = B^\top B$ to the full discrete Hodge-1 Laplacian (Paper 30 Prop. 2) — and the cubic term is nonzero, with a vertex proportional to ϵ^{abc}. The kinetic normalization is the standard one of the Wilson action's naive continuum limit,

$$\text{coeff} = \frac{1}{4N_c} \implies \tfrac{1}{8}\ (\text{SU}(2)), \quad \tfrac{1}{12}\ (\text{SU}(3)), \qquad (1)$$

checked numerically (Paper 30 Remark 1). Restricting the links to the maximal torus gives the compact $U(1)$ Wilson action (Paper 30 Prop. 3, checked to machine precision); this is a statement about the action, not the path integral, since Haar measure on SU(2) does not restrict to Haar measure on the torus, so compact $U(1)$ is a separate theory, solved on the scalar graph by the same factorization, $\langle \cos\theta_C \rangle = (I_1/I_0)^A$. The scope is explicitly bounded: this is a compact-graph construction, *not* a Yang–Mills mass-gap or confinement claim on $\mathbb{R}^4$, and the label graphs have no continuum limit.

c. Where the gauge reach stops. The non-abelian story does not continue to SU(3) as a *natural* matter-coupled theory. On the S^5 Bargmann–Segal lattice (the harmonic-oscillator analog of S^3) a pure-gauge Wilson SU(3) is well defined (the same $1/(4N_c) = 1/12$ coefficient and a Cartan reduction to a coupled $U(1) \times U(1)$ character are checked, Paper 30 Remark 1 and §4.1), but the natural matter sector cannot be coupled in the strict Wilson sense: the Clebsch–Gordan intertwiner obstruction re-appears at the matter level, and the m_ℓ-quotient is not a spectral discretization of $\mathbb{CP}^2$ (Paper 24). This is the gauge-side face of the Coulomb/HO asymmetry catalogued in Paper 24 [5]: $U(1)$ transfers universally, SU(2) is the natural choice because S^3 = SU(2), and SU(3) admits a pure-gauge Wilson theory but no natural matter coupling. The cumulative reading is that the natural gauge content of the construction saturates at $U(1) \times \text{SU}(2) \times \text{SU}(3)$—the structural reach, not a derivation of the Standard-Model gauge group, and consistent

Construction	Rules that hold
Scalar Fock graph	1 (triangle sparsity)
Native Dirac graph (rule B)	1, 4, 5, 7 (by edge definition)
Vertex model	1–5, 8; 7 for Dirac electrons (α_z); 6 not stated
Physical vertex	1, 4; 6 (Ward–Takahashi); 3 as representation theory

TABLE I. Paper 33's three-way census of eight selection rules (five standard, three specific to S^3). With the physical vertex rules 2, 5 and 8 fail and rule 7 (Furry) is undecided.

with the spectral-triple gauge analysis of Paper 32 [8].

d. Compact $U(1)$ on the Dirac Rule-B graph (Paper 30 §5). On the scalar graph the gauge theory factorizes into disjoint per-ℓ planar grids, so it is the exactly solvable two-dimensional theory. The cross-ℓ Dirac "Rule B" graph (the $E1$ dipole adjacency, $|\Delta\ell| = 1$) is different: every edge changes ℓ, the graph is connected, and its 4-cycles outnumber the independent cycles (44 against 11 at $n_{\max} = 2$) and close into 2-cycles of three plaquettes, so compact $U(1)$ does not factorize. At fixed diffusion scale its spectral dimension sits about 0.45 above the scalar graph's. Compact $U(1)$ on Rule B is tested against seven diagnostics (spectral dimension, Migdal–Kadanoff flow, Gaussian perimeter law, strong-coupling area law, a DeGrand–Toussaint monopole density falling roughly exponentially in β, full Monte-Carlo Wilson loops, and Polyakov loops on $G_B \times C_{N_t}$). All are compatible with compact $U(1)$ and none discriminates a dimension: no four-dimensional control was run, the monopole data end below the four-dimensional transition, the joint-fit area coefficient is consistent with zero at $n_{\max} = 2$, and the Polyakov loop is well defined through one extra cohomology class while its expectation vanishes by centre symmetry. Each link lies in many plaquettes (8.8 and 37.5 on average at $n_{\max} = 2, 3$), so the comparison class is mean-field or large-dimension lattice gauge theory, not a three-dimensional lattice.

III. QED ON S^3 AND THE GRAPH: SELECTION RULES AND PERIODS

The QED arc asks how much of perturbative quantum electrodynamics is visible on the combinatorial graph, before any continuum projection, and what period content the continuum spectral sums carry.

a. The selection-rule census (Paper 33). On the scalar Fock graph with a scalar (1-cochain) photon exactly one of the eight rules survives—the Clebsch–Gordan triangle sparsity of the discrete edge set. The physical vertex $\int_{S^3}\langle\psi_1, c(A)\,\psi_2\rangle$, Clifford multiplication by a coclosed 1-form between Camporesi–Higuchi eigenspinors, is fixed by Spin(4) representation theory and confirmed by two exact routes. For same-sign spinors it keeps the vertex model's window and adds $q = n_1+n_2+1$; for opposite signs it opens channels with n_1+n_2+q even; the ground state couples, also to itself, and the Ward–Takahashi identity holds exactly. Scored against it, triangle sparsity and Δm hold, the Ward identity holds in Ward–Takahashi form, the channel count holds as representation theory, the ground-state zero, E1 parity and the level triangle fail, and Furry's theorem is undecided. The vertex model's 1+6+1 partition, whose Dirac Furry score rests on a diagonal α_z identity, does not survive the physical vertex.

b. The perturbative period tower (Paper 28). The S^3 Dirac spectral sums (Camporesi–Higuchi spectrum $|\lambda_n| = n + \frac{3}{2}$, Hodge-1 photon, the vertex model's selection rule) produce odd zeta values and the level-4 content G, $\beta(4)$ through a spectral mechanism organized by operator order and a parity restriction of the spectrum. The spectral-zeta identities are statements about the sums as defined and stand independently of the vertex model:

- **T9 (two-term theorem).** The D^2 spectral zeta is a two-term π^{even} polynomial with rational coefficients at every integer s—no odd-zeta content there (the one-loop determinant $\zeta'_{D^2}(0) = 3\zeta_R(3)/(4\pi^2) + \frac{1}{2}\ln 2$ does carry it); operator order is the primary transcendental discriminant, conditional on the period conjecture.

- χ_{-4} **bridge (Theorem 3).** A spectral-zeta-to-Dirichlet-L identity, $D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}\big(\beta(s) - \beta(s-2)\big)$ with $\beta(s) = L(s, \chi_{-4})$, in which the parity of n acts as the Dirichlet character (that the vertex model's parity clause acts this way on internal lines is a reading of the model); proven, and verified symbolically and by PSLQ to 100 digits.

- **Realized-depth tower.** Loop order bounds the multiple-zeta depth but does not automatically mint new irreducibles; the realized depth grows strictly more slowly, $\leq k - 1$. The two-loop sunset constant reduces entirely, $S_{\min} \in \mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$ (depth ≤ 1); the three-loop chain $S^{(3)} = 40.6554\ldots$ first introduces a depth-2 period, $\zeta(5, 3)$, as its only constant beyond the single-zeta ring (its irreducibility to single zetas is conjectural).

On the finite graph these quantities are π-free rationals or algebraic numbers—the transcendentals enter only through the infinite spectral sums of the continuum projection. The vertex model's ground-state self-energy zero is not reproduced on the graph (the pendant-edge proposition gives $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max} \to 2$, Paper 33). A one-loop magnetic-moment ratio of 1.084 in the vertex model divides a coupling-free sum by the CODATA $\alpha/(2\pi)$; its closeness to $1 + 2/25$ is a numerical coincidence (Paper 28).

IV. THE FINE-STRUCTURE-CONSTANT OBSERVATION

The arc's most precise—and most carefully bounded—result is Paper 2's. The fine-structure constant approximately satisfies (to 8.8×10^{-8} relative; an Observation) the depressed cubic

$$\alpha^3 - K\alpha + 1 = 0, \qquad K = \pi(B + F - \Delta) = \pi\left(42 + \frac{\pi^2}{6} - \frac{1}{40}\right), \tag{2}$$

where the three coefficients are spectral data of the unit S^3: $B = 42$ is the degeneracy-weighted trace of the $SO(3)$ Casimir $l(l+1)$ over the scalar S^3 harmonics of the first three shells (an S^3 quantity, not an S^2 one: the S^2 trace over $l \leq 2$ is 36; the criterion $B/N = \dim S^3 = 3$ is chosen and, given it, fixes the cutoff $n_{\max} = 3$ uniquely), $F = \zeta(2) = \pi^2/6$ is the Dirichlet series of the shell degeneracy n^2 at $s = 4$, the maximum degree $d_{\max}$ of the packing [1] (equivalently half the spectral zeta of the circle at $s = 1$, its $k > 0$ modes), and $\Delta = 1/40$ is the reciprocal of the single-level Dirac degeneracy $g_3^{\text{Dirac}} = 2(k+1)(k+2)|_{k=3} = 40$ (Camporesi–Higuchi index). The physical root gives $\alpha^{-1} = 137.036011$, matching CODATA to a relative error of 8.8×10^{-8}; the weights, signs, prefactor and the cubic are chosen. In a combinatorial search over 1.92×10^9 comparable formulas ten match more closely (the formula ranks eleventh; $p = 5.2 \times 10^{-9}$ per formula), so the expected chance count is $Np \approx 10$ and within that search the formula is not unusual; the search was run after the formula was found and does not account for that.

The status of Eq. (2) must be stated exactly, and it is a **hard prohibition** of the framework that it be stated no more strongly. The combination rule $K = \pi(B + F - \Delta)$ is an **Observation**: an empirical numerical coincidence with structural support, *never* a derivation, conjecture, prediction, or theorem. Each ingredient has an *identified* spectral home—Paper 2's "Three homes" Proposition records that B, F, Δ live in two distinct sectors of S^3 (scalar and spinor) and are three different object types (finite trace, infinite Dirichlet series, single-level degeneracy); no single construction generating all three has been exhibited (an empirical record, not part of the Proposition)—but their *combination* is not derived. A multi-phase structural-decomposition program eliminated twelve single-mechanism derivations of the rule, and no derivation of the combination rule exists. The synthesis inherits this verdict verbatim: what is structural here is each spectral home; what is observational is that they sum approximately to α^{-1}/π (to 4.8×10^{-7} relative; 8.8×10^{-8} through the cubic's root).

V. THE SPECTRAL-ACTION GRAVITY SKETCH

Paper 51 records one new theorem and the continuum spectral-action results of Chamseddine and Connes [15] that it serves, on the round S^3 and on the Euclidean black-hole cigar. None of its results uses the GeoVac label lattice: they concern the continuum Camporesi–Higuchi spectrum and a finite-difference cone.

The new result is a convergence theorem (Paper 51, Theorem 3.1; one standard ingredient argued rather than proved): on a finite-difference discretization of the two-dimensional spinor cone of apex angle $2\pi\alpha$, the heat trace converges in C^1 in α near $\alpha = 1$, so the replica derivative, whose mode sum carries an extra weight $(j + \frac{1}{2})$, converges at the heat-trace rate. Numerically the replica combination converges to the continuum spinor tip value $1/6$, at order $O(a)$ in the radial spacing.

The continuum results are these. The spectral action on the round S^3 (and $S^3 \times S^1$) is two-term up to exponentially small corrections—Einstein–Hilbert plus a cosmological constant—and its content is Chamseddine and Connes's; Paper 51 gives an alternative proof through the spectral-zeta identity $\zeta_{\text{unit}}(-k) = 0$ for all $k \geq 0$ (via the Bernoulli identity $B_{2k+1}(3/2) = (2k+1)/4^k$; Paper 51, Theorem 4.1 and Cor. 4.2). It is spinor-bundle-specific (the scalar and transverse-traceless heat traces do not truncate; Theorem 6.1), and the absence of higher-curvature terms is a statement about the conformally flat $S^3 \times S^1$. Traced over the rank-4 spinor bundle, the action gives $G = 3\pi/(\phi(1)\Lambda^2)$ for a cutoff with Mellin moments $\phi(s)$; the replica derivative of the cigar's tip term, with the Dirac cone coefficient and the fermionic sign of the effective action, gives $S_{\text{BH}} = A\Lambda^2\phi(1)/(12\pi) = A/(4G)$ with the same G. The cigar is not conformally flat, and there the two routes agree at leading order in Λ; their agreement is the induced-gravity statement of Susskind–Uglum and Frolov–Fursaev, specialised to this spectrum, not a new result. The second variation of the action on the $(1,1)$ sector of $SO(4)$ cannot distinguish the diagonal-SU(2) spins $J = 0, 1, 2$ (Paper 51, Prop. 8.2), so it cannot by itself isolate a graviton. On $S^3 \times \mathbb{R}$, the Euclidean Einstein static universe, the two-term density has a formal extremum; the exact mode sum is monotonic in the radius.

The negatives are stated plainly:

- the continuum spinor cone tip term is t-independent ($1/6$), and the discrete per-t tip recovery is a crossing of that value at the lattice scale, not a converged extraction;
- at excess angle $\alpha > 1$ the fixed-spacing recovery plateau is a discretization artifact: as the radial mesh refines, the recovery climbs toward the plain continuation of the cone coefficient;
- the Newton-constant "factor of 2" between the action and entropy routes came from tracing over a

rank-2 bundle (or reading the cone coefficient per component); the rank-4 trace and the total-trace coefficient make the two agree;

- the end-to-end discrete S_{BH} on the substrate is not converged: it depends strongly on Λ and on the t-grid, because the substrate's tip profile peaks at a t fixed by the lattice, not by the cutoff.

The overall verdict is that the round-sphere spectral action supplies a two-term Einstein–Hilbert skeleton (Chamseddine–Connes; an alternative proof here), the cutoff function remains external calibration data, and the full discrete-substrate re-derivation of S_{BH} is an explicitly open research program.

VI. HONEST SCOPE AND POSITIONING

a. What is robust. The load-bearing results are structural and proven. Wilson loops on the scalar graph factorize exactly (the plaquettes form a cycle basis), the weak-coupling kinetic term is the curl form that completes the edge Laplacian to the Hodge-1 Laplacian, and the maximal-torus restriction of the SU(2) action is the compact $U(1)$ action to machine precision (Paper 30). The perturbative period results—the T9 two-term theorem, the χ_{-4} Dirichlet-L bridge, and the realized-depth tower with $\zeta(5,3)$ as the first depth-2 period (conjecturally irreducible)—are theorems (T9, χ_{-4}) and a Proposition (the realized-depth tower, PSLQ-identified) about the sums as defined. The spectral-action two-term exactness on S^3 is a theorem (Chamseddine–Connes; alternative proof in Paper 51). The physical vector-photon vertex rule is fixed by Spin(4) representation theory and confirmed by two exact routes, with the Ward–Takahashi identity exact (Paper 33). Each of the three α-ingredients B, F, Δ has an identified spectral home.

b. What is conditional or observational. The fine-structure combination rule $K = \pi(B + F - \Delta)$ is an **Observation**, not a derivation—the single most important tier statement in the arc. The multi-loop period statements are about sums defined with a vertex model, not the physical QED vertex, and the physical vertex's coupling normalization and gauge sector are owed. The Rule-B gauge theory is compatible with compact $U(1)$ and no dimension is discriminated. The gravity sector's continuum results are Chamseddine–Connes's and Susskind–Uglum's specialised to this spectrum; its new content is the replica-derivative convergence theorem, and the discrete-substrate program is open.

c. What was tried and did not work. The arc is bounded by clean negatives. SU(3) has no natural matter-coupled lattice gauge theory—the Clebsch–Gordan intertwiner obstruction caps the natural gauge reach (a face of the Coulomb/HO asymmetry). The scalar graph photon misses seven of the eight selection rules because it carries no angular quantum numbers. The vertex model's ground-state zero, the plaquette "recovery" of it, the α_z Furry identity, the 1+6+1 partition, the anomalous-moment ratio's Parker–Toms match, and the Rule-B confinement readings are withdrawn or rescoped. Twelve single-mechanism derivations of the α combination rule were eliminated. The gravity sector contributes four recorded negatives—the per-t tip recovery as a crossing, the excess-angle finite-spacing artifact, the factor of 2 traced to the spinor rank, and the unconverged end-to-end discrete entropy—each kept on the record so it is not re-derived.

d. Positioning. Read as a whole, the Group 5 arc is a structural map of how far the spectral geometry of S^3 reaches into electromagnetism, the gauge sector, and gravity, with the boundary between forced structure and free coupling drawn explicitly at every step. The natural gauge constructions, the QED period tower of the spectral sums, and the two-term round-sphere action are the structural assets; the fine-structure constant, the QED multi-loop constants, and the gravity cutoff are the calibration side. The discrete and continuous descriptions are dual, with no ontological priority asserted for either. The arc is not a derivation of α, not a derivation of the Standard-Model gauge group, and not a theory of quantum gravity; it is a disciplined account of which structures the framework forces and which numbers it must be told.

[1] J. Loutey, "Counting Angular Momentum Labels by Isotropic Packing," GeoVac Paper 0 (2026).

[2] J. Loutey, "The Fine Structure Constant as a Spectral Coincidence on the Three-Sphere," GeoVac Paper 2 (2026). [The combination rule $K = \pi(B + F - \Delta)$ is recorded strictly as an Observation; see §IV.]

[3] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation on a Scale-Invariant Label Lattice," GeoVac Paper 7 (2026).

[4] J. Loutey, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," GeoVac Paper 18 (2026).

[5] J. Loutey, "The Bargmann–Segal Lattice: π-Free Discretization of the Harmonic Oscillator on S^5," GeoVac Paper 24 (2026).

[6] J. Loutey, "Periods of Dirac Spectral Sums on S^3," GeoVac Paper 28 (2026).

[7] J. Loutey, "Wilson Lattice Gauge Theory on the GeoVac Label Graphs: Exact Factorization on the Scalar Graph and Compact $U(1)$ on the Dirac Rule-B Graph," GeoVac Paper 30 (2026).

[8] J. Loutey, "The GeoVac Spectral Triple," GeoVac Paper 32 (2026).

[9] J. Loutey, "Selection Rules of the Vector-Photon Vertex on S^3: the Physical Vertex, the Vertex Model and the

Graph," GeoVac Paper 33 (2026).
[10] J. Loutey, "GeoVac as a Two-Layer Framework: Projection Taxonomy and the Empirical Matches Catalog," GeoVac Paper 34 (2026).
[11] J. Loutey, "The Leading One-Loop Hydrogen Lamb Shift: Assembly, a Convention Erratum, and Coulomb-Sturmian Bethe Logarithms," GeoVac Paper 36 (2026).
[12] J. Loutey, "State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on S^3," GeoVac Paper 38 (2026).
[13] J. Loutey, "A Convergent Discrete Replica Derivative on the Spinor Cone, with the Two-Term Camporesi–Higuchi Spectral Action on $S^3 \times S^1$," GeoVac Paper 51 (2026).
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[15] A. H. Chamseddine and A. Connes, "The Spectral Action Principle," Commun. Math. Phys. **186**, 731 (1997).